

University of Kentucky Pediatric Stroke Algorithm*

Inclusion Criteria: Age between 30 days to < 18 years old
AND focal neurological deficit with onset < 24 hours

No history of seizures with consistent post-ictal weakness as the only deficit

If the patient has sickle cell, consider transfer to the PICU for emergent apheresis after completing algorithm



Primary inpatient team, nursing, and ED stroke screening assessment: **BE-FAST**

B: Balance (ask about changes in balance or coordination)

E: Eyes (ask about vision changes or complaints)

F: Face (ask the patient to smile and look for asymmetry)

A: Arms and legs (ask the patient to lift the arms/legs for ten seconds and look for drift or weakness)

S: Speech (ask the patient to state his/her name and address)

T: Time (ask and document when the symptoms were noticed and when the patient was last well)

If one or more BE-FAST criteria are positive and inclusion criteria are met, ED provider, nurse, or primary inpatient team activates stroke alert (call 35200) and orders basic labs; e.g., Fingerstick, UDS, CBC, CMP as clinically indicated



If no BE-FAST criteria are positive, consider a routine Child Neurology consult



Intravenous thrombolytic Candidate Requirements:

Age: 2+ years

Last known well: less than 4.5 hours ago

Thrombectomy Candidate Requirements:

Age: 2+ years

Last known well: less than 24 hours ago



Yes to one or both set(s) of requirements

CT head without contrast to be ordered only by Neurology **and** if there is concern for hemorrhage. Primary team to contact Child Neurology to arrange MRI head with and without contrast within 8-10 hours (ROUTINE) as needed; include time frame needed in order. Patient is to be NPO, have an IV, and have sedation arranged at **Child Neurology** discretion. Neurology will add MRV head with contrast (ROUTINE) as clinically indicated.



If venous sinus thrombosis or hemorrhage are present, STAT Neurosurgery consult.



Confirmed stroke patients ≥15 will be admitted to the Adult Stroke service. Primary team will transfer to PICU if the patient (<15) is a thrombolytic or thrombectomy candidate. Otherwise, disposition per primary team in consultation with Neurosurgery or Child Neurology as appropriate; Adult Stroke Neurology will follow as needed.

Studies to be ordered at Neurology or Neurointerventionalist discretion if the patient is a thrombolytic and/or thrombectomy candidate:
CT Head without contrast (STAT)
CTA Head/Neck (STAT)
CT Brain Perfusion (STAT)



If there is hemorrhage, Neurology will call Neurosurgery and obtain MRV head with contrast and MRI head with and without contrast (priority and sedation to be determined by Neurosurgery and/or **Child Neurology**). If the patient is a thrombolytic and/or thrombectomy candidate, Neurology will speak with the **Child Neurology attending** and/or Neurointerventionalist on call and administer thrombolytic and/or activate thrombectomy alert at their discretion.



Epic Pediatric Stroke Order Set:
ED Peds Stroke Alert
COM PED ICU Stroke Protocol

* The University of Kentucky Pediatric Stroke Guidelines have been developed by UK Healthcare and are primarily intended for use within our institution and, therefore, may need to be modified for use at other institutions. These guidelines do not indicate an exclusive course of action. The guidelines are not intended as a substitute for the independent clinical judgment of the treating provider, as the guidelines do not account for individual variation among patients. The UK Pediatric Stroke Guidelines are available on Careweb.